

ESPRIT · 2026

SOFTWARE ENGINEERING

PEAXIS

A Modular AI-Powered Hiring Operating System

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AI-Assisted Recruitment

Explainable AI

Multi-Tenant SaaS

Modular Architecture

1 INTRODUCTION & CONTEXT

Presentation Plan

An engineering defense, from problem to results.

01 Industry & Problem

The recruitment landscape today, and the engineering problems within it.

02 Internship Context

The environment, technologies, and engineering exposure behind this work.

03 Existing Solutions & Gap

Where current recruitment platforms fall short.

04 Methodology & Requirements

Engineering process, functional goals, and quality constraints.

05 Proposed Solution & Architecture

PEAXIS: modules, technology stack, and system architecture.

06 Results & Conclusion

Engineering achievements, lessons learned, and future work.

2 INDUSTRY & PROBLEM

Recruitment in the Digital Era

Four forces are reshaping how organizations hire.



Application volume

Online job boards and easy-apply flows multiply applications per opening.



Recruiter workload

The same recruiter teams review a growing volume of candidates per role.



Candidate expectations

Applicants expect fast feedback, transparency, and a clear application status.



AI adoption in HR

Hiring teams increasingly expect AI support in screening and decision-making.

- Hiring has become a higher-volume, higher-expectation process — this is the environment in which recruitment software now has to operate.

2 INDUSTRY & PROBLEM

Current recruitment challenges

Engineering problems behind the hiring process, not just symptoms.

42 days

Avg time-to-hire
SHRM 2024

75%

Recruiters overloaded
LinkedIn Talent Report

60%

Candidates report no feedback
Indeed Survey 2023



Manual screening

Reviewing every CV by hand does not scale with rising application volume.



Keyword-based ATS

Legacy filters match exact keywords and miss well-qualified, non-standard CVs.



Inconsistent evaluation

Criteria vary by reviewer, with limited traceability of how a decision was reached.



Candidate experience

Long application forms and no status feedback push qualified talent away.

Cost of a bad hire: averages **\$14.9K**, while strong candidates accept competing offers within days.

Regional gap: many SMEs in the MENA region still rely on spreadsheets, as legacy ATS tools are costly and poorly localized.

3 INTERNSHIP

Internship Context

This project was carried out during an internship at **Prospecter**, an AI-powered B2B sales prospecting platform, as a Software Engineer.



Full-Stack Engineering

Production Next.js and NestJS codebases, with tenant-scoped data models.



AI Integration

LLM pipelines for parsing, classification, and structured generation.



Platform Engineering

Docker deployments, async queues, and semantic search infrastructure.

Scope: PEAXIS is an independent codebase and product, built with academic rigor and the production standards observed during the internship.

SOFTWARE ENGINEER

AI INTEGRATION

SOLO CONTRIBUTOR

4 EXISTING SOLUTIONS & GAP

Competitive Analysis

How existing recruitment platforms cover core ATS, candidate experience, and explainability.

CAPABILITY	GREENHOUSE	LEVER	WORKABLE	ASHBY	HIREVUE	EIGHTFOLD AI	PEAXIS
Core ATS	✓	✓	✓	✓	✗	partial	✓
Candidate Portal	basic	basic	basic	✗	✓	✗	✓
Explainable AI	✗	✗	partial	✗	partial	✓	✓
Integrated Ecosystem	✗	✗	partial	partial	✗	✗	✓
Tenant Isolation	✓	✓	✓	✓	✓	✓	✓

Observation: few platforms combine core ATS workflows, a candidate-facing portal, and explainable AI scoring in a single, SME-accessible product.

4 EXISTING SOLUTIONS & GAP

The gap: integrated, explainable AI



AI black boxes

Recommendations need traceability: requirements, cited evidence, and recruiter review.



Fragmented tooling

Specs, pipelines, calendars, and notes live in separate tools.



No candidate intelligence

Tools track applications but rarely guide decisions or interviews.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



This gap defines the project's objective

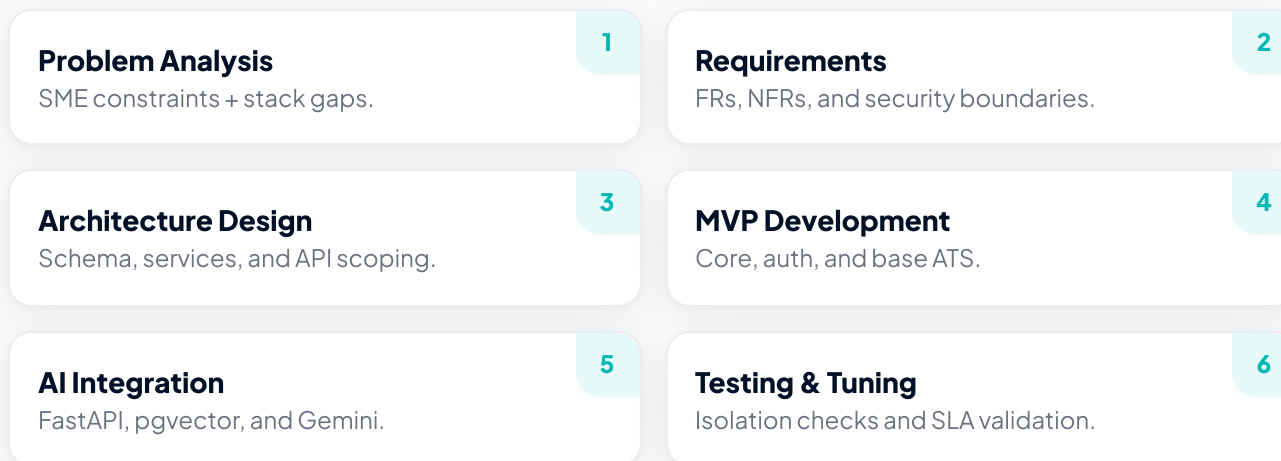
Explainable scoring, integrated workflows, and a candidate-facing experience in a single system.

5 METHODOLOGY & REQUIREMENTS

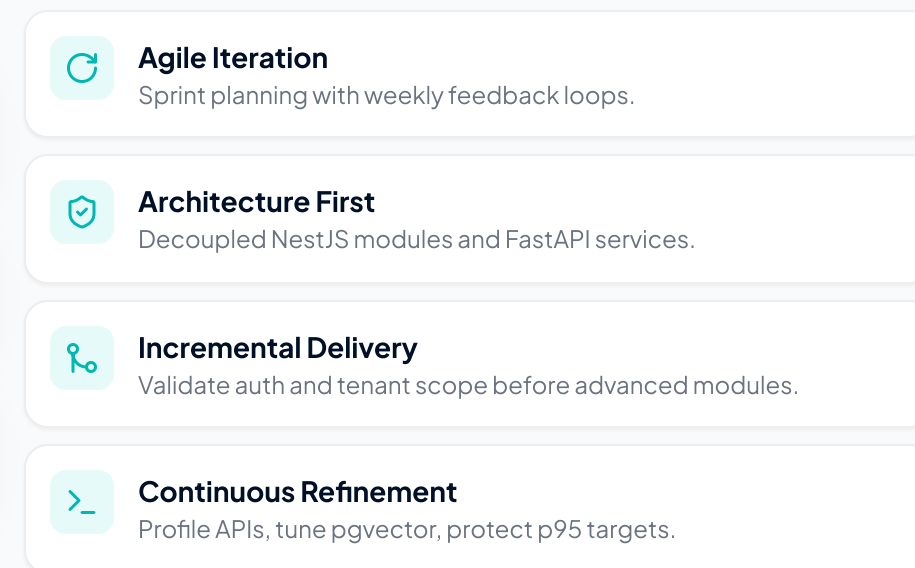
Engineering Methodology

The process used to move from problem to production-grade system.

PROJECT ENGINEERING PIPELINE



EXECUTION STRATEGIES



5 METHODOLOGY & REQUIREMENTS

Core platform requirements

FR-01 to FR-05: the foundation for a production-grade recruitment SaaS.



FR-01

Authentication & Security

- Argon2 password hashing
- Google + LinkedIn OAuth2
- 15m access + 7d refresh JWT



FR-02

Multi-Tenant Architecture

- Business-scoped queries
- X-Business-ID enforcement
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise
- JwtAuthGuard + PlanGuard



FR-04

Job Management

- CRUD + rich filters
- Semantic search via pgvector
- Automatic embeddings



FR-05

Application Pipeline

- Applied to Hired/Rejected
- Stage transition history
- AI score at submission

5 METHODOLOGY & REQUIREMENTS

AI & analytics requirements

FR-06 to FR-10: the intelligence layer above traditional ATS workflows.



FR-06

AI Match Scoring

- Skill-overlap engine
- 0–100 score + explanation
- Strength/weakness breakdown



FR-07

CV Parsing

- PDF / DOCX / TXT upload
- Configured provider extraction
- Durable candidate-profile parsing



FR-08

Candidate Onboarding

- Apply with CV
- Profile completeness score
- Candidate embeddings



FR-09

Interview Scheduling

- Interview CRUD + type
- Application-linked schedule
- Interviewer assignment



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking

5 METHODOLOGY & REQUIREMENTS

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods; Redis queues absorb spikes.



OWASP aligned

NFR-02 **Security**

Argon2, JWT rotation, OAuth2, tenant scoping, OWASP checks.



p95 < 200ms

NFR-03 **Performance**

Redis caching + BullMQ keep heavy work off requests.



99.5% SLA target

NFR-04 **Availability**

SDK retries, job retries, and Docker health checks.



Full audit trail

NFR-05 **Explainability**

Scores include match, label, skills, gaps, and explanation.



Microservices

NFR-06 **Modularity**

NestJS modules, FastAPI service, decoupled frontends.

6 PROPOSED SOLUTION

PEAXIS — Modular AI Hiring Operating System

Three integrated modules. One ecosystem, one data layer, every stakeholder served.



Operational Backbone

PEAXIS Core

Auth, billing, tenant governance, roles, and accounts.



Recruiter Execution

PEAXIS Hire

ATS pipeline, scoring, summaries, interviews, and analytics.



Talent Acquisition

PEAXIS Jobs

Semantic search, CV parsing, match preview, and easy apply.

BUSINESS ADMIN

PEAXIS Core

Governance, billing, teams, settings

RECRUITER / HR TEAM

PEAXIS Hire

Pipelines, scoring, analytics, collaboration

CANDIDATE / TALENT

PEAXIS Jobs

Search, match preview, easy apply

• All modules share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

6 PROPOSED SOLUTION

PEAXIS Core

The shared business foundation every module relies on.



Identity & Access

Authentication, tenant isolation, and role-based permissions.



Billing & Plans

Subscription management through Stripe.



Team & Governance

Seats, settings, and account-level policy.

| One control center for tenancy, access, and billing across every module.

JWT RS256

MULTI-TENANT

STRIPE BILLING

6 PROPOSED SOLUTION

PEAXIS Hire

The core ATS for AI-assisted recruiter execution.



Workspace & Pipeline

Kanban pipeline, candidate profiles, and offer management in one workspace.



Recruiter Intelligence

AI match scoring, evidence review, and funnel analytics support each decision.



Collaboration & Flow

Scheduling, team approvals, and shared notes keep hiring decisions auditable.

One audit-ready interface for tracking, collaboration, and evaluation.

6 PROPOSED SOLUTION

PEAXIS Jobs

The candidate portal for discovery, matching, and low-friction applications.



Job Discovery

Semantic search and filters surface roles beyond keyword matching.



Application UX

AI CV parsing pre-fills profiles for a low-friction apply flow.



AI Matching & Insights

Skill-gap feedback and alerts help candidates target the right roles.

Replaces opaque applications with clear matching insights and faster apply flows.

6 PROPOSED SOLUTION

AI ownership boundaries

AI assists with structured inference; the platform owns records and rules, while recruiters retain hiring authority.



NestJS API

Owns authorization, tenant scope, business records, deterministic assessment rules, and persistence.



Dedicated worker

Executes durable AI work, applies retry rules, and reconciles queued work without blocking HTTP requests.



FastAPI service

Performs parsing, embeddings, citation-bound classification, and supported content generation only.



PostgreSQL + pgvector is authoritative for records, snapshots, and vectors.



Redis + BullMQ carries queues, locks, heartbeats, and non-authoritative caches.

NESTJS DECIDES BY RULES

FASTAPI INFERS

RECRUITER DECIDES TO HIRE

7 ARCHITECTURE & TECHNOLOGIES

Tech Stack — Four Layers

Production-grade architecture. Modular, scalable, and designed for real-world hiring at scale.

Frontend

Core, Hire, Jobs, and Landing apps with shared React/TypeScript patterns

Next.js 16

React

Tailwind

TypeScript

API

NestJS + Prisma: 14 modules, 60+ endpoints, one data authority

NestJS

Prisma

JWT

TypeScript

AI Service

Stateless inference for parsing, classification, embeddings, and structured generation

FastAPI

Gemini / Azure

Python

Redis

Database

Authoritative data + pgvector retrieval, Redis, and one durable BullMQ queue

PostgreSQL

pgvector

Redis

BullMQ

MULTI-TENANT

Every query scoped to businessId

ASYNC PROCESSING

BullMQ + background workers

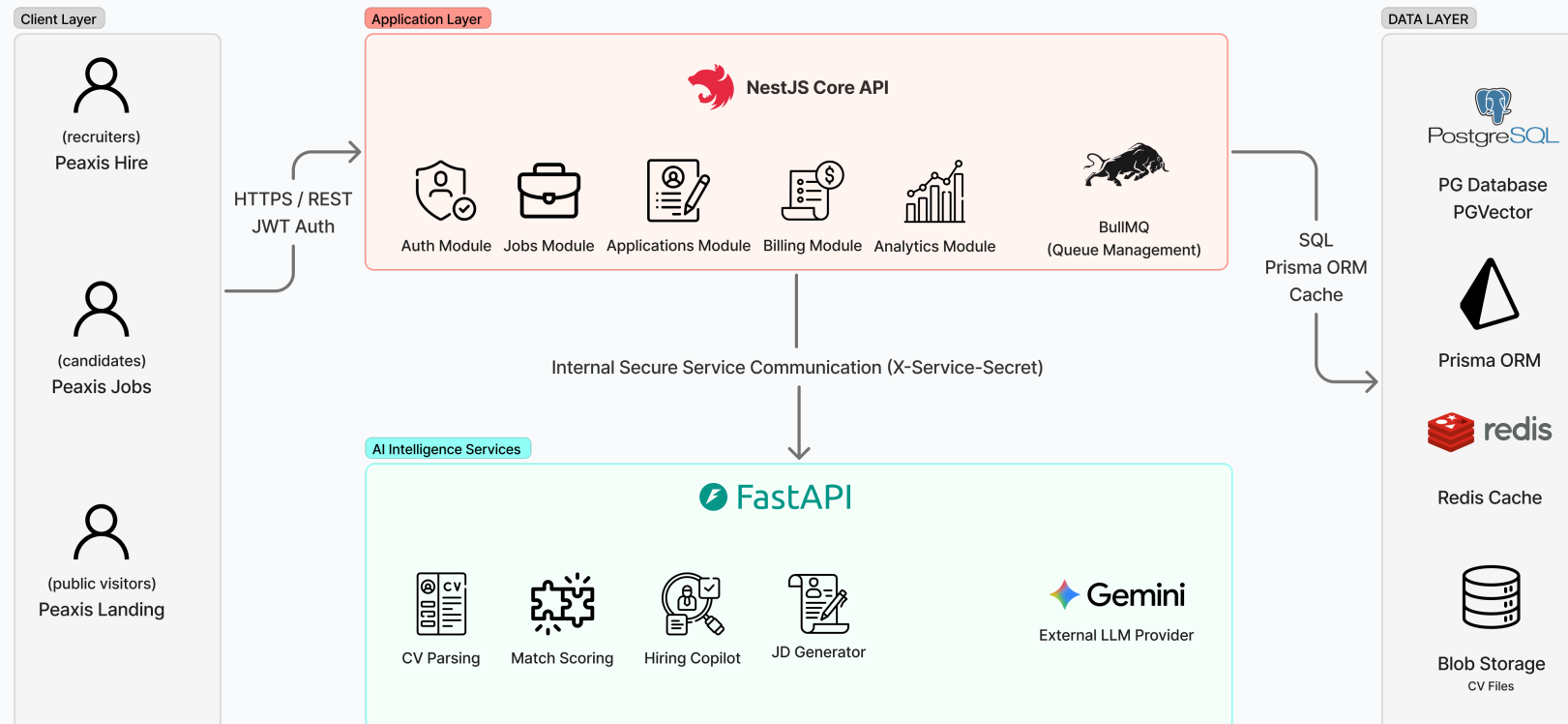
RETRIEVAL

pgvector supports job discovery and cited evidence

7 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

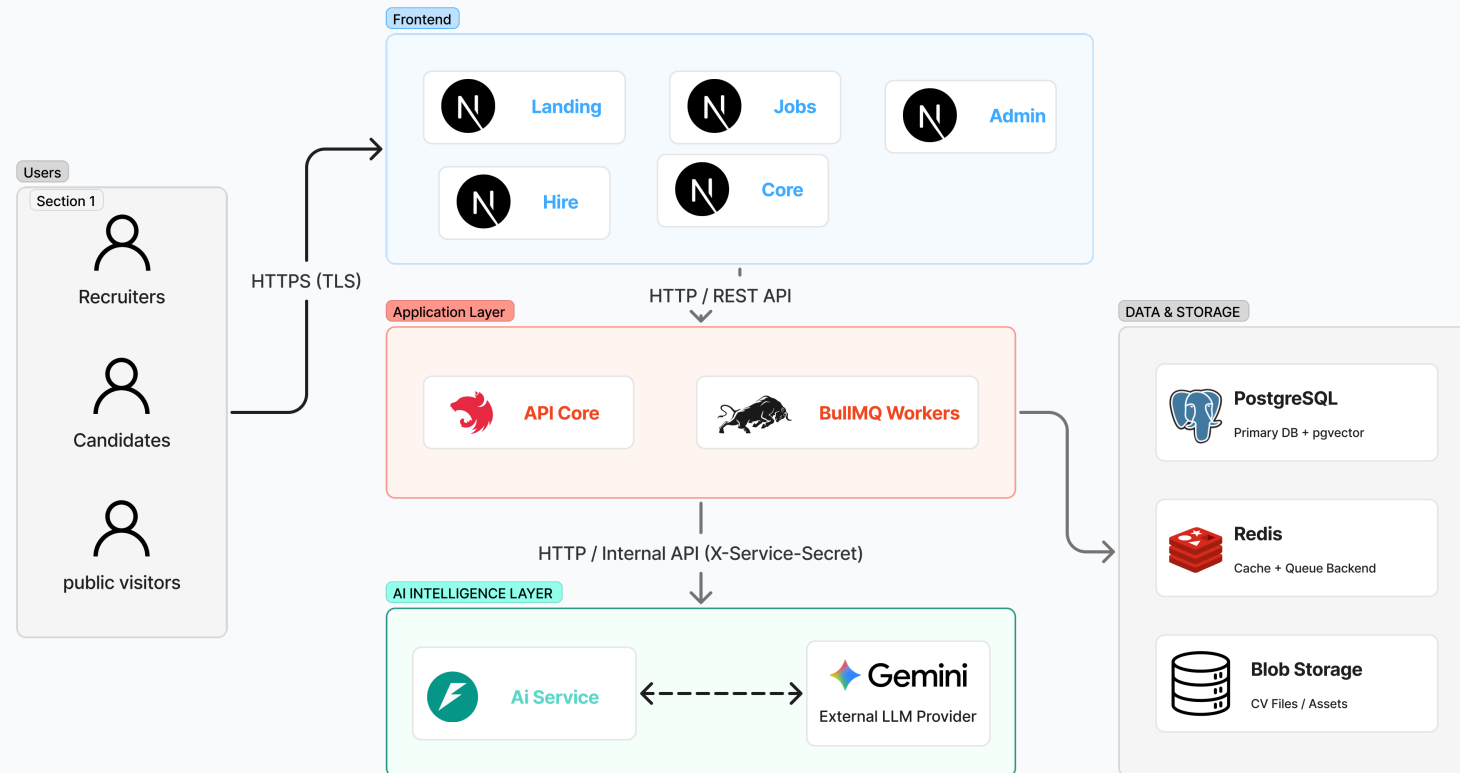
PEAXIS connects its client applications, core platform, AI services, and data layer through clear service boundaries.



7 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

The deployed topology separates frontend applications, platform services, AI inference, and persistent storage.



8 ENGINEERING DEEP DIVE

Architecture Decisions

The stack was selected for delivery risk, maintainability, and production behavior.

Modular monolith

vs microservices

Domain stays cohesive; less distributed complexity.

Separate AI service

FastAPI

Python ML stack isolated from core API load.

Next.js apps

3 frontends

Server rendering plus rich client interactions.

PostgreSQL

Prisma

Relational workflows with strong consistency.

BullMQ

async

Heavy AI work moves outside request lifecycle.

pgvector

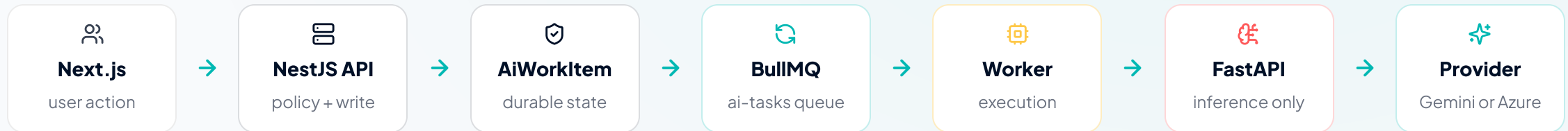
retrieval

Retrieval for job discovery and cited evidence.

8 ENGINEERING DEEP DIVE

AI Runtime Architecture

NestJS orchestrates durable work; FastAPI performs inference; PostgreSQL remains the source of truth.



Synchronous request path

Validate, authorize, persist business state, create work state, and return a processing or accepted response.

Asynchronous worker path

Call FastAPI, persist parse/embedding/assessment outputs, and update the durable work and application state.

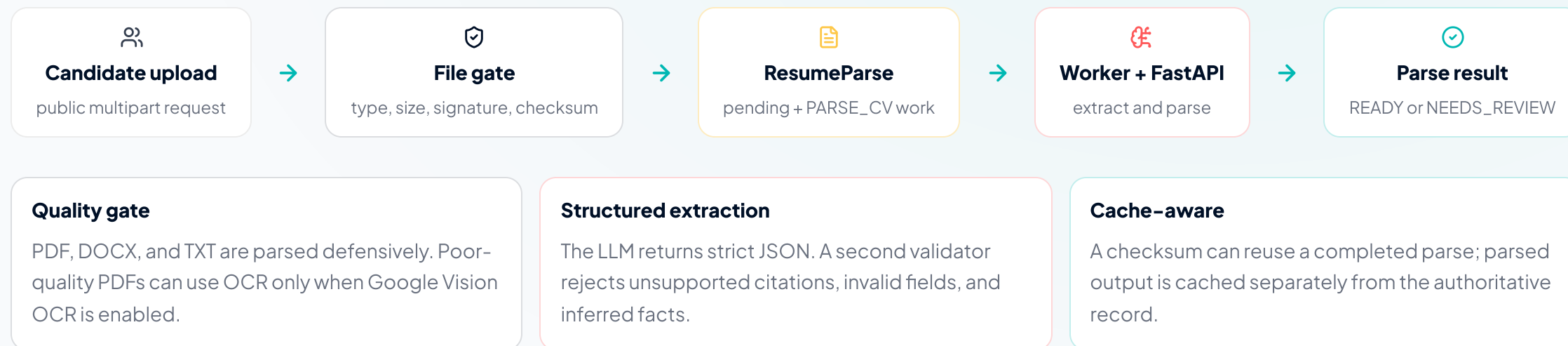
Return path: the worker writes results to PostgreSQL; Redis is used for queues, locks, heartbeats, and caches—not authoritative AI state.

8 ENGINEERING DEEP DIVE

CV Parsing Pipeline

The résumé is validated and parsed asynchronously before its structured facts become candidate evidence.

1. INTAKE AND DURABLE PARSING



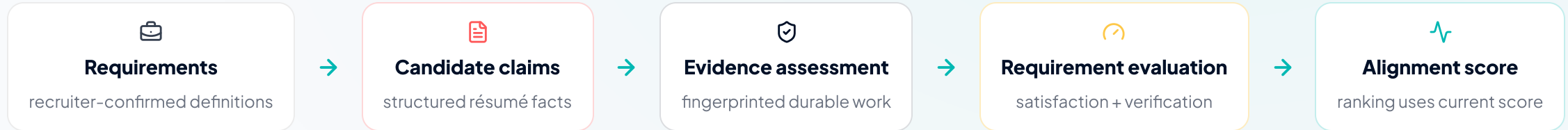
2. CANDIDATE CONFIRMATION AND RETRIEVAL PREPARATION



8 ENGINEERING DEEP DIVE

Evidence-Based Matching Engine

The model can classify cited evidence; the backend determines satisfaction, verification, alignment, and the reviewable result.



AI responsibility

When direct evidence is absent, retrieve up to eight similar candidate chunks, then classify only the supplied citation set.

Backend responsibility

Exact matching, experience-month calculation, score weights, satisfaction, and verification states are deterministic NestJS rules.

Human responsibility

Recruiters define requirements, inspect evidence, and decide the candidate outcome. No AI hiring decision exists.

The current alignment and ranking score are the same persisted calculation—not a separate opaque candidate similarity model.

How Matching Is Computed

The assessment is a deterministic evidence algorithm. Vectors and the LLM only help discover or classify a bounded citation.

1. Find candidate evidence

Canonical token and simple stem matching compare each requirement with source-cited claims. The fallback searches displayable résumé summary, headline, experience, and project text. Experience requirements merge overlapping date intervals into relevant months.

2. Use vectors only when needed

For unresolved COMPETENCY, RESPONSIBILITY, or TOOL_SYSTEM requirements, FastAPI embeds the requirement. pgvector retrieves only this candidate's chunks at distance ≤ 0.55 (maximum 8). The classifier sees at most 12 claims plus retrieved chunks and may cite only one supplied source.

3. Apply deterministic requirement states

No evidence, mandatory
REQUIRES_VERIFICATION

Evidence but years unmet
PARTIALLY_SATISFIED

No evidence, preferred
UNKNOWN

Evidence + policy satisfied
SATISFIED

4. Persist explainable results

An **EvidenceAssessment** stores input snapshots and the score. Each **RequirementEvaluation** retains satisfaction, constraints, and positive cited evidence.

CURRENT GENERAL:V1 SCORE

$$\text{score} = \text{round} \left(\frac{\sum (\text{weight} \times \text{state value} \times \text{evidence credit})}{\sum (\text{weight})} \right)$$

State value: SATISFIED = 100, PARTIALLY_SATISFIED = 60, all other states = 0. The current ranking score equals this alignment score; no separate opaque cosine ranking is used.

required 0.55

experience 0.25

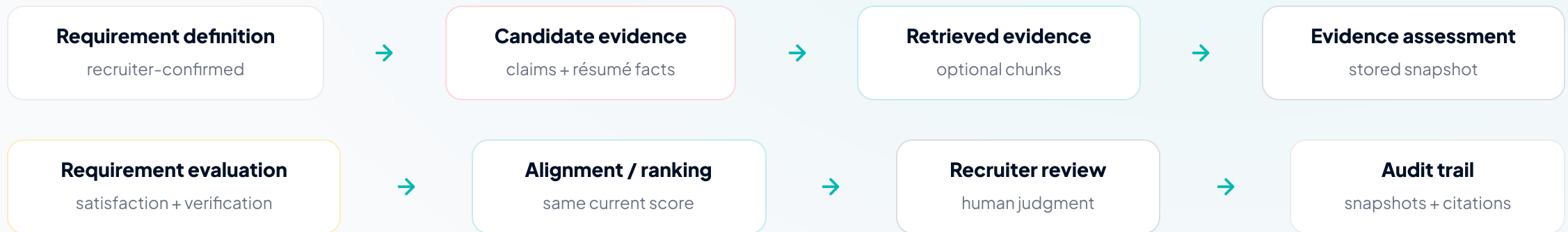
preferred 0.10

TRANSFERABLE × 0.60

8 ENGINEERING DEEP DIVE

Explainable AI

Every assessment is traceable from a recruiter-confirmed requirement to cited candidate evidence and a deterministic evaluation.

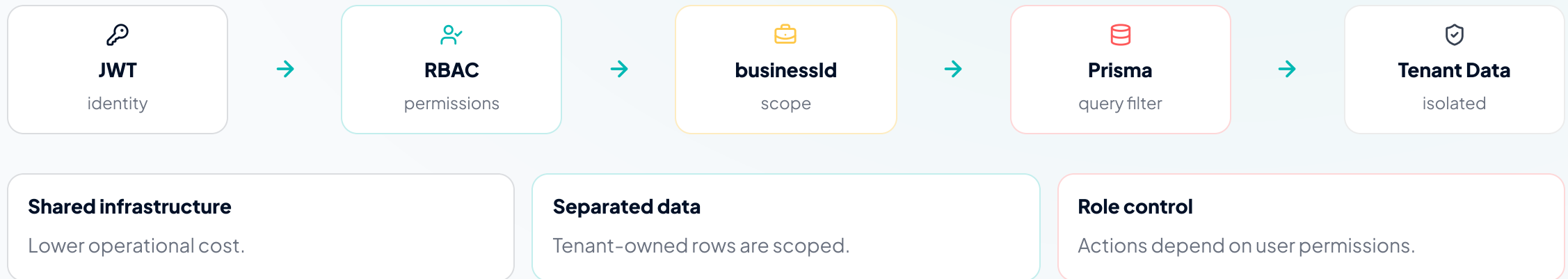


The LLM can classify supplied citations; NestJS applies scoring and verification rules. The system never makes a hiring decision.

8 ENGINEERING DEEP DIVE

Multi-Tenant Architecture

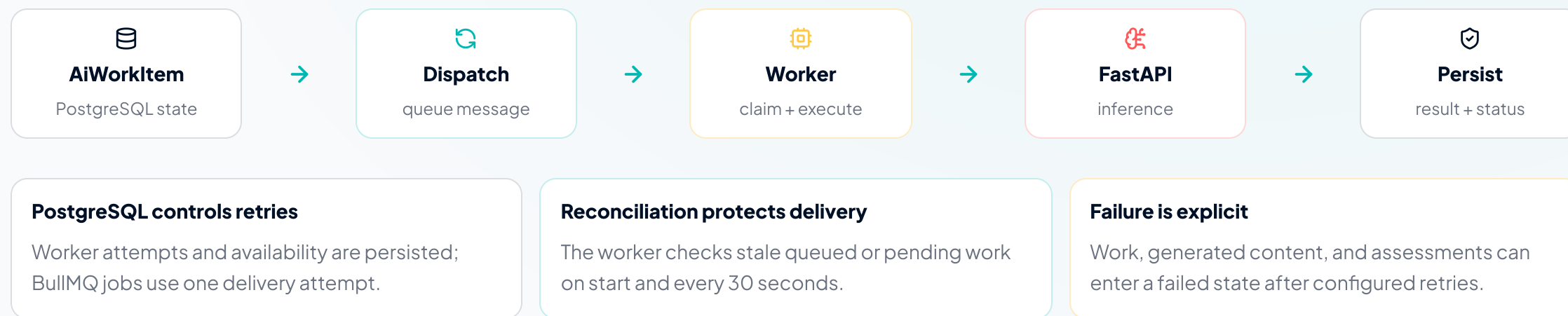
One shared platform, separated business data, and request-level tenant isolation.



8 ENGINEERING DEEP DIVE

Background Processing

PostgreSQL owns durable work state; BullMQ delivers execution to a dedicated worker without blocking the request.



No dedicated dead-letter queue is implemented

8 ENGINEERING DEEP DIVE

Security Architecture

Implemented controls span identity, tenant scope, file intake, and service boundaries; known gaps remain explicit.



JWT

Authenticated API access



Membership

Active tenant membership in protected flows



businessId

Tenant-scoped domain records



Rate limits

Public CV and FastAPI request throttling



File gates

Size, MIME, extension, and signature checks



Service secret

FastAPI internal-call authentication



Provider PII

Raw CV content is sent for inference



Visibility gap

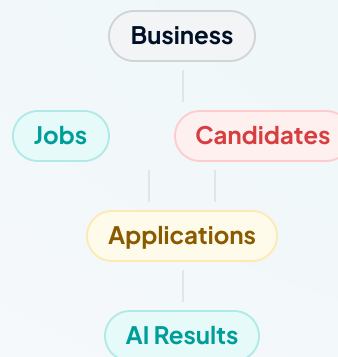
Candidate recommendation access needs remediation

Pre-production hardening: enforce active membership for generated content and define provider retention, consent, and PII redaction policy.

8 ENGINEERING DEEP DIVE

Database Design

PostgreSQL models the hiring workflow, while pgvector adds semantic search.



Indexes

businessId, status, relations, search keys.

Vectors

Retrieval support for public job search and cited evidence.

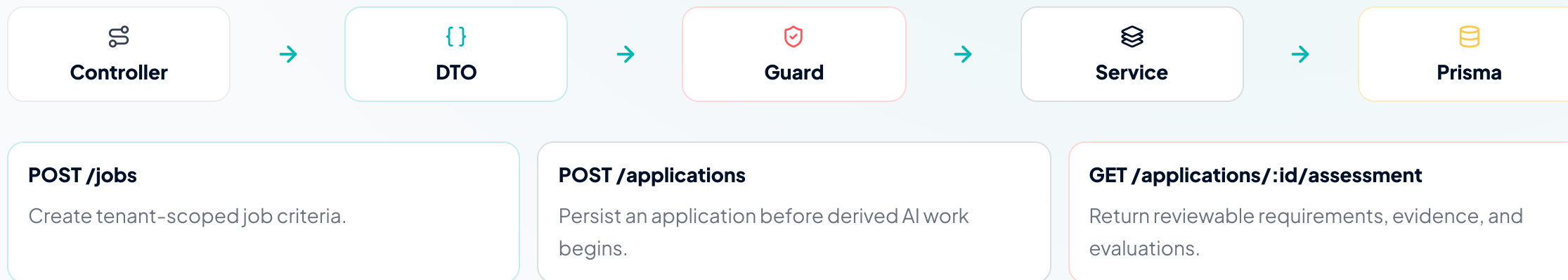
Audit

Stored AI evidence and workflow history.

8 ENGINEERING DEEP DIVE

REST API Design

NestJS keeps HTTP contracts, policy, business logic, and persistence separated.



AI Models & Routing

One configured provider—Gemini or Azure OpenAI—is selected at startup; there is no silent provider fallback.

Feature	Model route	Reason
CV parsing	Configured provider	Structured résumé extraction
Evidence classification	Configured provider	Only supplied citations are classified
Embeddings	Configured embedding model	Job search and candidate evidence retrieval
Generated content	Configured provider	Drafted, reviewable content
Assessment score	NestJS + PostgreSQL	Deterministic rule application

Provider selected at startup

FastAPI rate limit: 60/min per IP

Structured JSON where supported

Raw résumé content reaches the provider

8 ENGINEERING DEEP DIVE

Performance Optimizations

The implemented system favors bounded retrieval, cache reuse, and durable work over synchronous AI latency.



Redis cache

Bounded parse, embedding, and selected-generation reuse; never the source of truth.



Durable worker

Slow inference happens outside HTTP requests with task-specific retry policies.



Hybrid job search

PostgreSQL full-text search merges with pgvector semantic results.



Chunk retrieval

Evidence retrieval is constrained to the candidate and returns only a small cited set.



Provider retries

The inference client retries transient provider/network failures.



Graceful search

If query embedding fails, public job search returns keyword results with degraded status.

Known limit: pgvector is enabled, but the baseline migration does not create an ANN vector index.

Technical Challenges & Solutions

The system has deliberate safeguards and documented limitations that remain visible for an academic implementation review.

Challenge	Solution	Outcome
AI latency	Durable work + worker retries	Responsive request path
CV reliability	File gates + quality gate + optional OCR	READY or NEEDS_REVIEW state
Grounding	Citation-bound evidence classification	No free-form evidence invention
Explainability	Snapshots, evaluations, and citations	Auditable assessment trail
Provider outage	Retries and explicit failure states	Recoverable, observable processing
Known access gap	Candidate recommendation visibility needs remediation	Documented before production

Roadmap & Future Work

The next milestones harden the current implementation before widening AI capability.

NEAR-TERM

Access control

Enforce active membership for all AI-generated content and correct recommendation visibility.

Atomic delivery

Commit application and assessment work atomically with a transactional outbox pattern.

Reviewer policy

Make overrides affect an explicit versioned assessment policy and explainability.

MID-TERM

FinOps coverage

Route every chargeable AI call through operations and credit ledger reconciliation.

Prompt controls

Centralize prompt construction and test every user-controlled input surface.

Vector lifecycle

Add stale-chunk cleanup and measure before introducing an ANN vector index.

LONG-TERM VISION

Governed expansion

Expose only supported FastAPI helpers through tenant-authorized NestJS contracts.

Data governance

Define provider retention, PII classification, consent, and operational redaction.

Results & Conclusion



Engineering Achievements

- A full-stack multi-tenant SaaS: Core workspace, Hire ATS, and Jobs portal
- Evidence-based assessment: cited requirements, reviewer overrides, and an audit trail
- Durable AI processing: queued work outside the request path via a dedicated worker



Architectural Achievements

- Clear separation of concerns across frontend, API, worker, and inference layers
- Tenant isolation enforced at the schema and request level, not retrofitted
- Semantic retrieval with pgvector, and a containerized, independently scalable runtime



Lessons Learned

- Multi-tenant guards must be designed in from the first schema, not added later
- A stateless inference boundary keeps business state and authority in one place
- Bounded AI — provider limits, credits, and mandatory review — keeps risk manageable



PEAXIS demonstrates that AI-assisted recruitment can remain reviewable, tenant-safe, and operationally bounded — with people retaining authority over hiring decisions.

Thank you Questions?

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